Gabriel Klyatis

gabrielklyatis@gmail.com | LinkedIn | +1 (445) 207 0474

Education

Expected: May 2028 **University of Pennsylvania**
MSE, Robotics

2024 **Tel Aviv University** (GPA: 83.95/100)
B.Sc., Electrical and Electronic Engineering

Professional Experience

Software Engineer, Core | Magic Leap (Google) Sept 2024 - Aug 2026

- Design, develop, and maintain embedded software on microcontroller platforms.
- Research and implement system components on SOC (Android) and MCU (RTOS) platforms.
- Analyze and optimize the display pipeline between the Application Processor (AP) and the MCU, reducing latency and improving power efficiency for constrained hardware.

Intern Software Engineer, Device Security | Magic Leap May 2023 - Sept 2024

- Researched and implemented custom device security solutions for Magic Leap's platform and device architectures, enhancing the stock Android OS security flow.
- Collaborated with security experts to implement advanced security tools, contributing to the team's overall objectives - signing and validating builds, kernel, hardware and additional components of the company's device.
- Developed and enhanced secure, high-quality applications like IrisID, while modifying stock Android apps (e.g., keymaster, gatekeeper) for the device's secure processor. Additionally, implemented security tools using test-driven development and improved code reusability and system robustness.

Skills

Programming Languages:	C, C++, Python
Operating Systems:	Embedded Linux, Android, RTOS
Low-Level & System Software:	Device Drivers, Services, MCU firmware (Pigweed)
Hardware Debugging:	JTAG, MCU Link, Trace32, Saleae Logic Analyzer
Protocols & Interfaces:	I2C, SPI, UART, GPIO, MIPI DSI, Network Protocols (TCP, IP, TLS, etc.)
Development Tools & Workflows:	Git, Gerrit, Bazel, GDB, Linux Command Line
Computer Science Fundamentals:	Data Structures, Algorithms, Multithreading, IPC
Languages:	English (Fluent), Hebrew (Native), Russian (Fluent)

Projects

B.Sc. Final Project | *TCP/IP Stack, Advanced Computer Communications Laboratory*

Designed and developed a modular TCP/IP stack in C++, consisting of 15,000+ lines of code for system architecture-based implementation and 1,700 for testing. The stack supported hundreds of future electrical engineering and computer science students in using the Advanced Computer Communications Laboratory.